

Effect of constrictivity of gel/capillary pores in concrete on chloride ions migration

Yafen Zhang

Open University of China

Ruonan Liu

North China Electric Power University

Ruicheng Zhang

North China Electric Power University

Xiaoyu Yan

yanxiaoyu@ouchn.edu.cn

Open University of China

Zhuo Zhao

Beijing University of Technology Beijing

Yulong Zhu

North China Electric Power University

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Abstract

Corrosion of steel bars due to chloride ions in seawater migration into reinforced concrete (RC) lining is a major factor affecting the lifetime of subsea tunnels. To improve the safety assessment of existing subsea tunnels, a coupled hydraulic-mechanical-chemical (H-M-C) model is proposed to simulate the chloride ions migration process with hydrostatic pressure in the RC lining of subsea tunnels for obtaining the long-term distribution of chloride ions in the RC lining more accurately. In the H-M-C coupled model, the volume fluid fraction and convection velocity obtained from the bidirectionally coupled hydraulic and mechanic analysis are unidirectionally considered in the analysis of convection, diffusion, and adsorption of chloride ions in the RC lining. In addition, to consider the influence of concrete microscopic pores (e.g., gel pores and capillary pores) size on chloride ion migration, the classic expression of the effective diffusion coefficient is modified by considering a constrictivity factor that varies nonlinearly with the microscopic pore size. Results indicate that in the diffusion zone, the concentration of chloride ions significantly increases with increasing gel/capillary pores radius (r^{peak}), leading to a rapid non-linear decrease in the service time of the RC subsea tunnel. Afterward, to more clearly ascertain the sensitivity of the effects of constrictivity of gel/capillary pores in concrete on chloride ion migration, the sensitivity analyses are carried out on four sets of parameters (i.e., saturated permeability, van Genuchten parameters, initial saturation, and binding capacity parameters). The results of the sensitivity analyses suggest that the effects of capillary pores radius (r^{peak}) on the penetration process of chloride ions in the concrete lining of subsea tunnels are more sensitive to the initial saturation (S_e) than the vG parameters (a and m). Furthermore, with the increase of capillary pores radius (r^{peak}), the sensitivity of the chloride ion penetration to capillary pores radius the under different chloride binding conditions is increasing.

1. Introduction

Reinforced concrete (RC) lining is the most important supporting structure of subsea tunnels, and its durability is an important factor related to tunnel safety, operating costs, and service life (Qu et al., 2007). Compared to mountain tunnels, subsea tunnels are usually large high-risk projects due to the high water pressure and unlimited water recharge (Niu et al., 2022). Once the local damage to the subsea tunnel occurs, the huge amounts of unstoppable water gushing or bearing capacity loss of the lining structure of the subsea tunnel will lead to a large number of casualties inside the tunnel (Zhang et al., 2012), and even the whole tunnel would also be scrapped (Chen et al., 2022a), which is undoubtedly a terrible disaster. In the past few decades, the corrosion damage of steel bars has been considered to be the primary factor causing countless structural failures and has brought high costs to society in repairing, monitoring, and/or replacing those failed structures (Guan et al., 2011; Berrocal et al., 2016). On the other hand, seawater provides a high concentration of chloride ion environment. The chloride ion attack and carbonation are the most prominent factors inducing the degradation of subsea tunnels (James et al., 2019). The harm of chloride ions to the safety of RC lining is multifaceted, which eventually manifests itself as corrosion of reinforcement (Hsieh, 2015). The corrosion-induced cracking or spalling of the cover reduces the bond strength in the steel and concrete interface (Zhao et al., 2013), thereby reducing the

service life of RC structures in the ocean (Zheng et al., 2023). Therefore, as an RC structure in a marine environment, subsea tunnels are facing a long-term threat of chloride ions, and more attention should be paid to the monitoring and prediction of chloride ions migration depth in the subsea tunnel lining.

Based on the chemical particle diffusion in solution, many scholars (Zhang et al., 2016; Zhang et al., 2019; Chen et al., 2022b; Gaylarde and Ortega-Morales, 2023) are committed to giving a more accurate chloride ion migration model. For example, Zhang et al. (2016) proposed a diffusion-advection theoretical model of chloride ions migration in subsea concrete tunnels based on the basic parameters of the concrete (e.g., porosity, saturated hydraulic conductivity, chloride diffusion coefficient, etc.) with two water-cement ratios determined by laboratory tests. Chen et al. (2022b) proposed a new nonlinear chloride transport model by considering the time-depending surface chloride concentration and concrete parameters, as well as the seepage velocity varies with the radial direction of the tunnel. Duzy et al. (2023) presented the comparison of results obtained from chloride migration tests conducted on three Alkali-Activated Concrete (AAC) mixtures and mentioned that in a marine zone, the chloride ions affect the properties of structures. Szveda et al. (2023) presented a comparison of the protective properties of three concretes of similar composition on the effect of chloride ions and found that the type of cement influences the rate of chloride diffusion in the concrete. While it is worth noting that when concrete structures are subjected to external environmental influences (e.g., high ground stress or high water pressure), the microscopic pore structure in the concrete material is prone to change (Yeh, 2013), which causes constrictivity of the concrete microscopic pore to vary. However, the effect of the constrictivity of the concrete microscopic pores on chloride diffusion is usually neglected in the traditional chloride migration model, i.e., the Millington and Quirk (MQ) model (Millington and Quirk, 1959). In the MQ model, the effects of concrete microscopic pores on chloride ions migration is only determined by the volumetric water content and porosity of the concrete. However, Szveda et al. (2023) found that there seems to be no correlation between the porosity of concrete and the number of pores (pore size) according to the results of porosity analysis of concrete cores using X-ray computed microtomography. This means that the pore structure of concrete could not be clearly described by concrete porosity alone, and a separate variable must be considered in chloride ions simulations to give a more detailed description of the effect of pore structure. Consequently, to eliminate this shortcoming of the traditional chloride migration model (MQ model), Zhang et al. (2019a) modified the classical expression for the effective diffusion coefficient by considering a constrictivity factor based on numerical modeling of unsaturated concrete, but in their study, the constrictivity factor is considered as a constant and does not consider the nonlinear effect of changes in microscopic pore structures. Therefore, it is necessary to consider not only the influence of the concrete porosity but also the influence of microscopic pore size (pores number) in the simulation of chloride ions migration.

To reflect the influence of non-linear constrictivity on the migration process of chloride ions in unsaturated concrete, this paper proposes a coupled hydraulic-mechanical-chemical (H-M-C) model based on Richard's equations, porous solid mechanics, and Fick's second law. In the H-M-C coupled model, the volume fluid fraction and convection velocity obtained from the bidirectionally coupled

hydraulic and mechanic analysis are unidirectionally considered in the analysis of convection, diffusion, and adsorption of chloride ions in the reinforced concrete lining.

2. Hydraulic-Mechanical-Chemical (H-M-C) Coupling Model.

The migration process of chloride ions into concrete involves convection, diffusion (Wang et al., 2017), dry-wet cycle, and relative humidity (Zhang et al., 2019b). In subsea tunnels, the inner side of the lining is a dry environment, and the outer side is a saturated environment with high water pressure. As an unsaturated porous material, the subsea tunnel lining involves many problems such as strain, seepage, chloride ions diffusion, and adsorption. Therefore, the process of chloride ions migration into the concrete lining of subsea tunnels is regarded as a fluid-solid-solute three-physical field coupling problem. To give a more realistic solution, a coupled hydraulic-mechanical-chemical (H-M-C) model is established in this paper to simulate the chloride ions migration process with hydrostatic pressure in the reinforced concrete (RC) lining of subsea tunnels. In the H-M-C coupled model, the volume fluid fraction and convection velocity obtained from the bidirectionally coupled hydraulic and mechanic analysis are unidirectionally considered in the analysis of convection, diffusion, and adsorption of chloride ions in the RC lining as indicated in Fig. 1.

The mechanic governing equation for fluid-solid coupling is as follows:

$$G \nabla^2 u_i + \frac{G}{1-2\nu} u_{j,ji} + F_i - \alpha_p p_{,i} = 0$$

$\underbrace{\hspace{10em}}_{Solidphase} \quad \underbrace{\hspace{10em}}_{Fluidphase}$

1

where, G is the shear modulus (Pa); u_{ij} and u_{jji} are the displacement tensors (m); α_p is the Biot's effective stress parameter which depends on the compressibility of the concrete ($\alpha_p = 1 - K'/K$); K' is bulk modulus of the porous medium and K is the bulk modulus of the solid grains (Pa); ν is the drained Poisson's ratio; p is pore pressure (Pa); F_i is body force (Pa).

Richard's equation (Richard, 1931) is widely used to characterize the seepage of water in unsaturated porous media:

$$\underbrace{\left[C_m + S_e S \right]}_{SolidPhase} \frac{\partial H_p}{\partial t} + \nabla \bullet \left[\underbrace{-k(\theta)}_{Solidphase} \bullet \nabla (H_p + D) \right] = 0$$

2

where, the specific moisture capacity C_m relates variations in fluid volume fraction θ to the matric head (i.e., $C_m = \partial \theta / \partial H_p$). The storage coefficient S addresses storage changes due to compression and expansion of the pore spaces and the water when the porous medium is fully wet, which sets $S = \rho_f g (\chi_p + \theta \chi_f)$. Here, ρ_f is the fluid density (kg/m³), g is the acceleration due to gravity (m/s²), while χ_p and χ_f the compressibility of solid particles (1×10^{-8} m·s²/kg) and fluid (4.4×10^{-10} m·s²/kg), respectively. S_e is the effective saturation and $k(\theta)$ is unsaturated hydraulic conductivity (m/s), defined as $k(\theta) = k_r \cdot k_s$. Here, k_r denotes the relative hydraulic conductivity and k_s is hydraulic conductivity (m/s) in saturated concrete. The pressure head H_p (m) is a dependent variable (matric potential driven by hydrostatic pressure and capillary pressure) and D is the vertical elevation (m).

Van Genuchten (1980) adopted the Mualem model (Mualem, 1976) to propose a simple water retention curve to describe the relationship between θ and k_r in unsaturated porous media.

$$k_r = \left(\frac{\theta - \theta_r}{\varphi - \theta_r} \right)^l \left[1 - \left(1 - \left(\frac{\theta - \theta_r}{\varphi - \theta_r} \right)^{1/m} \right)^m \right]^2$$

3

The relationship between porosity (ϕ), solid stress (σ), and pore pressure (p) is given by the following equation (Zimmerman et al., 1986; Cui and Bustin, 2005; Detournay and Cheng, 1993).

$$\varphi = 0.8 - (0.8 - \varphi_0) \exp \left\{ -\frac{1}{K'} \left[\left(\bar{\sigma} - \bar{\sigma}_0 \right) + (p - p_0) \right] \right\}$$

4

The mass conservation equation describing the migration of chloride ions through concrete can be expressed as:

$$\frac{\partial}{\partial t} \theta C_f + \frac{\partial}{\partial t} C_{bound} = -\nabla \bullet J_{ion}$$

5

where, θ is the volume fluid fraction; C_f is the free chloride ion concentration (mol/m³) of pore solution; J_{ion} is the flux of chloride ion (mol/(m³·s)); t is the time (s); C_{bound} is the amount of bound chloride ion per unit volume (mol/m³ of concrete), defined as $C_{bound} = \rho_b \cdot C_p$, where ρ_b is the bulk density of concrete (kg/m³); C_p is the mass of binding chloride ion per dry unit weight of concrete (mol/kg for concrete). The relationship between C_p and C_f is non-linearly described by chloride binding isotherm k_p (m³/kg) (i.e., $k_p = \partial C_p / \partial C_f$).

Tang and Nilsson (1993) and Thomas et al. (2012) pointed out that k_p obeys the Freundlich isotherm at high free chloride concentrations as shown in Eq. (6).

$$C_p = \alpha C_f^\beta, k_p = \frac{\partial C_p}{\partial C_f} = \alpha \cdot \beta \cdot C_f^{\beta-1}$$

6

where, α and β are the constants. The effective diffusion coefficient for chloride ion transfer by diffusion mechanism in porous media is denoted by D_e . The MQ model (Millington and Quirk, 1959) has been widely adopted to estimate the D_e in porous media. In the MQ model, D_e is described by pore structures (i.e., volume fluid fraction, θ , and tortuosity factor, τ_{MQ}) and diffusion coefficient of chloride ion in water (D_0):

$$D_e = \frac{\theta}{\tau_{MQ}} D_0$$

7

where, $\tau_{MQ} = \theta^{7/3} \cdot \phi^2$ is the bending coefficient, representing the pore structure, and ϕ is the total effective porosity of the concrete.

In total, the governing partial differential equation (PDE) of the chloride ions migration can be combined as follows:

$$(\theta + \rho_b k_p) \frac{\partial C_f}{\partial t} + C_f \frac{\partial \theta}{\partial t} = \underbrace{\frac{\partial}{\partial x} \left(D_e \bullet \frac{\partial C_f}{\partial x} \right)}_{diffusion} + \underbrace{\frac{\partial}{\partial x} (k_r k_s \bullet \frac{\partial h}{\partial x} \bullet C_f)}_{convection}$$

8

In solving the partial differential equation (PDE) for chloride ion transfer, another model proposed by Nakarai et al. (2006) to calculate D_e (referred to as the NI model) is as follows.

$$D_e = \theta \bullet \frac{\delta}{\tau_{NI}} \bullet D_0$$

9

where, δ is the constrictivity factor, which is used to describe the reduction effect of the pore connectivity and electric changes on the walls of the micro-pores during mass transport as shown in Fig. 2.

The constrictivity factor, δ , is related to the radius of the largest distributed pores (r^{peak}). The relationship between δ and r^{peak} is,

$$\delta = 0.395 \tanh [4 (lgr^{peak} + 6.2)] + 0.405$$

10

The bending coefficient τ_{NI} in NI model is defined as:

$$\tau_{NI} = -1.5 \tanh [8.0 (\varphi - 0.25)] + 2.5$$

11

According to the NI model, the traditional MQ model is modified by considering the constrictivity factor (δ):

$$D_e = \theta \bullet \frac{\delta}{\tau_{MQ}} \bullet D_0$$

12

Therefore, in the modified MQ model, the bending degree of the microscopic pores of concrete is characterized by the bending coefficient (τ_{MQ}), and the size of the pores is characterized by the constrictivity factor (δ).

3. Properties of Materials

The thick concrete lining is 0.3 m. The chloride ion concentration inside the concrete was initially assumed to be zero. The initial saturation of the concrete is 0.8 and the external water pressure is 1.5 MPa. In this paper, the concentration of chloride ions in seawater is taken as 546 mol/m³. The density of concrete is 2400 kg/m³, and the diffusion coefficient of chloride ions in water (D_0) is taken as 1.484×10⁻⁹ m²/s. The water-cement ratio of the tunnel lining concrete is 0.5. The bulk modulus of the concrete is taken as 17.5 GPa, and the shear modulus is taken as 13 GPa. The other input parameters required for the calculations are listed in Table 1.

Table 1
Input parameters used in the H-M-C coupled model (w/c = 0.5).

Category	Parameter	Value
Mechanical parameters (Nguyen et al., 2017)	Density, ρ_b (kg/m ³)	2400
	Porosity, φ	0.12
	Shear modulus, G (GPa)	13
	Bulk modulus, K' (GPa)	17.5
	Saturated permeability, k_s (m ²)	3×10^{-21}
Hydraulic parameters (Poyet et al., 2011; Pang et al., 2015)	vG parameter, a (m ⁻¹)	2.8×10^{-4}
	vG parameter, m	0.417
	vG parameter, l	0.5
	Initial water content, θ_0	0.096
	Residual water content, θ_r	0
Chemical parameters (Thomas et al., 2012)	Diffusion coefficient of chloride ions in water, D_0 (m ² /s)	1.484×10^{-9}
	Binding parameter, α	8.51
	Binding parameter, β	0.32

The pore types with the greatest impact on the constrictivity of concrete are capillary and gel pores. Figure 3 shows the number of gel/capillary pores with different sizes of three concrete samples measured by Gargepuram et al. (2021). It shows that most of the pore radius is distributed in the range of 0.05-10 μm and a small amount of pore radius distribution in more than 10 μm . In this paper, the values of the gel/capillary pore radius r^{peak} are set from 0.1 μm to 1.0 μm by investigating the effect of constrictivity of gel/capillary pores in concrete on chloride ions migration. Besides, the sensitivity of the chloride ion migration to gel/capillary pores constrictivity in different cases such as the different saturated permeability (k_s), the different initial saturation degrees (S_e), and different vG parameters (a and m), and the different binding parameters (α and β) are also discussed by parametric analysis.

4. Results and Discussions

4.1 Effect of gel/capillary pores constrictivity of concrete on chloride ions migration

In solving the partial differential equations (PDEs) for chloride ion transfer, the effective diffusion coefficients of chloride ions are determined by the traditional MQ model, the modified MQ model, and the NI model, respectively. The results of the chlorine ions concentration profiles in concrete under different models with radius 1×10^{-7} m is plotted in Fig. 4. From the comparison results of the chloride ions concentration profiles, it can be seen that the concentration of chloride ions calculated by the traditional MQ model (red lines) is significantly larger than that calculated by the NI model (blue lines) and the modified MQ model (black lines) in the same period. In the subsea tunnel RC lining, the thickness of the outer protective layer ranges from 50 mm to 100 mm (Song and Zhou, 2012; He et al., 2020; Guo et al., 2021). To quantitatively evaluate the service life of the subsea tunnel, the critical chloride concentration of reinforcement corrosion is assumed to be 55 mol/m^3 (Liu and Cai, 2022). It suggests that the calculation results of the traditional MQ model are untrue and too large. After modifying the MQ model by considering the pore constrictivity factor (δ), the concentration of chloride ions calculated by the modified MQ model (black lines) agrees well with that calculated by the NI model (blue lines), and both of them meet the design life of 50 years. Therefore, the results indicate that the constrictivity factor (δ) has a certain effect on the migration of chloride ions in concrete. Consequently, in this paper, the effect of the constrictivity factor (δ) of gel/capillary pores on the migration process of chloride ions in the concrete lining of subsea tunnels will be discussed.

To check the effectiveness of the proposed H-M-C coupled model, a comparative analysis is performed between the numerical simulation results and the experiment data. Three kinds of plain concrete with different fly ash content and different gel/capillary pores radius (r^{peak}) are taken as test samples. The input parameters used in the validation model are listed in Table 2. The remaining mechanical and hydraulic parameters are referred to in Table 1.

Table 2
Input parameters used in the validation model (adopted from Liu et al. (2020)).

ID	Chloride percentage at the concrete surface, $C_s/\%$	Gel/capillary pores radius, $r^{peak}/\times 10^{-7}\text{m}$
PCFA0	0.8210	4.0
PCFA15	0.7614	3.75
PCFA30	0.7187	3.5

The content of chloride ions simulated by the traditional MQ model (blue lines), the NI model (red lines), and the modified MQ model (black lines) are plotted in Fig. 5. The black scatters in Fig. 5 are the experimental results measured by Liu et al. (2020) in concrete after being immersed in a 16.5% sodium chloride solution for 35 days. The comparative analysis between the numerical results and experiment data indicates that the NI model has the highest calculation accuracy, followed by the modified MQ model. The calculation accuracy of the traditional MQ model is very low. This means that porosity alone

cannot accurately describe the migration of chloride ions in concrete structures, and the size of micropores needs to be introduced as an independent variable.

4.2 Effect of gel/capillary pores constrictivity on the lifetime of RC concrete

To explore whether the micro-diameter of concrete will significantly affect the migration of chloride ions, Fig. 6 represents the variation of the chloride ions concentration for different models with different gel/capillary pores radius (r^{peak}). Results indicate that with the increase of the gel/capillary pores radius, the increase of chloride concentration in the concrete is very large at the location of the steel bar ($d = 100$ mm). When the gel/capillary pores radius (r^{peak}) is 1×10^{-7} m, the chloride ion concentration calculated by the NI model is 0.755 mol/m³, 8.196 mol/m³, 19.610 mol/m³, 31.187 mol/m³, and 41.637 mol/m³ respectively after 10 years, 20 years, 30 years, 40 years and 50 years. While, when the gel/capillary pores radius (r^{peak}) is 3×10^{-7} m, these values are increased to 40.348 mol/m³, 84.265 mol/m³, 135.699 mol/m³, 180.063 mol/m³, and 216.373 mol/m³ respectively, meaning that the size of micropores has a significant influence on the migration of chloride ions in concrete.

Figure 7(a) shows the time-dependent concentration of chloride ions at the location of the steel bar ($d = 100$ mm). It can be seen that when the gel/capillary pores radius (r^{peak}) of the concrete is relatively small (e.g., 1×10^{-7} m and 3×10^{-7} m), the concentration of chloride ions increases almost linearly with time; while when the gel/capillary pores radius (r^{peak}) of the concrete is relatively large (e.g., 5×10^{-7} m, 7×10^{-7} m, and 1×10^{-6} m), the concentration of chloride ions increases non-linearly with time, and the increase is more rapidly in the first 10 years than after the 10 years. Figure 7(b) shows the lifetime of the RC subsea tunnel, which is defined by the initial corrosion time of steel bars at a chloride ion concentration of 55 mol/m³. Figure 7(b) suggests that the service time of the RC subsea tunnel rapidly decreases as the gel/capillary pores radius (r^{peak}) increases on a non-linear curve. Furthermore, due to the high requirements on the service life of undersea tunnels, the gel/capillary pores radius (r^{peak}) of the concrete is recommended to be controlled below 1×10^{-7} m by adding admixtures. In this case, by appropriately increasing the thickness of the protective layer, the service time of the RC subsea tunnel can be effectively extended. Conversely, once the gel/capillary pores radius (r^{peak}) is larger than 1×10^{-7} m, the service time of the RC subsea tunnel will be greatly shortened. At this time, the increase in the thickness of the protective layer will no longer have a significant protective effect on the steel bars in the RC subsea tunnel.

4.3 Sensitivity of the chloride ion migration to gel/capillary pores constrictivity under different hydraulic conditions

To check the sensitivity of gel/capillary pores constrictivity under different hydraulic conditions, Fig. 8 shows a comparison of the chloride ion concentration profiles produced at different values of saturated

permeability, k_s (i.e., $k_{s-1} = 3 \times 10^{-22} \text{ m}^2$, $k_{s-2} = 3 \times 10^{-21} \text{ m}^2$, and $k_{s-3} = 3 \times 10^{-20} \text{ m}^2$) after 20 years. The results show that in the diffusion zone, the concentration of chloride ions significantly increases with increasing gel/capillary pores radius (r^{peak}), while in the convection zone, the concentration of chloride ions under k_{s-1} , k_{s-2} , and k_{s-3} conditions are significant different in the small gel/capillary pores radius (e.g., $1 \times 10^{-7} \text{ m}$), but are much close in the large gel/capillary pores radius (e.g., $5 \times 10^{-7} \text{ m}$, $7 \times 10^{-7} \text{ m}$, and $1 \times 10^{-6} \text{ m}$), meaning that the increase of the chloride ions concentration is controlled by the saturated permeability (k_s) in the convection zone.

To investigate the sensitivity of gel/capillary pores radius (r^{peak}) on van Genuchten (vG) parameter and initial saturation, Fig. 9 represents the effect of different moisture characterization curves vG-1 ($a_1 = 2.8 \times 10^{-4} \text{ m}^{-1}$, $m_1 = 0.417$) and vG-2 ($a_2 = 2.8 \times 10^{-4} \text{ m}^{-1}$, $m_2 = 0.3371$) and the initial saturation degree S_e on the diffusion process of chloride ions in the subsea tunnel. Results show that the concentration of chloride ions increased significantly with the increase of the initial saturation. At the same initial saturation, the curves of vG-1 and vG-2 have a high degree of overlap, while at different initial saturations, the difference between the two curves is large, which indicates that the effects of gel/capillary pores radius on the migration process of chloride ions in the concrete lining of subsea tunnels are more sensitive the initial saturation S_e than the vG parameters (a and m).

4.4 Sensitivity of the chloride ion migration to gel/capillary pores constrictivity under different chloride binding conditions

Three kinds of chloride binding relationships (abbreviated CBR) are used, i.e., CBR-1 ($\alpha_1 = 8.51, \beta_1 = 0.32$), CBR-2 ($\alpha_2 = 6.71, \beta_2 = 0.68$), CBR-3 ($\alpha_3 = 9.82, \beta_3 = 0.58$). Figure 10 shows the concentration profiles of chloride ions calculated by different binding parameters (α and β) in 20 years. It can be seen that the diffusion process is affected by the change in binding parameters and this effect mainly reaches the maximum in the middle of the diffusion zone. With the increase of gel/capillary pores radius (r^{peak}), the sensitivity of the chloride ion migration to gel/capillary pores radius under different chloride binding conditions is increasing.

5. Conclusions

In this paper, to improve the safety assessment of existing subsea tunnels, based on Richard's equations, porous solid mechanics, and Fick's second law, a coupled hydraulic-mechanical-chemical (H-M-C) model is proposed to simulate the chloride ion migration process with hydrostatic pressure in the RC lining of subsea tunnels for obtaining the long-term distribution of chloride ions in the RC subsea tunnel more accurately. The findings from this study can be outlined as follows:

(1). The size of the gel/capillary pores in the concrete has a significant effect on the migration of chloride ions in the RC subsea tunnel. Therefore, it is necessary to consider a new independent variable, i.e., the pore constrictivity factor (δ), that varies nonlinearly with the gel/capillary pores radius (r^{peak}) in the traditional Millington and Quirk (MQ) model.

(2). When the gel/capillary pores radius (r^{peak}) of the concrete is relatively small (e.g., 1×10^{-7} m and 3×10^{-7} m), the concentration of chloride ions increases almost linearly with time; while when the gel/capillary pores radius (r^{peak}) of the concrete is relatively large (e.g., 5×10^{-7} m, 7×10^{-7} m, and 1×10^{-6} m), the concentration of chloride ions increases non-linearly with time, and the increase is more rapidly in the first 10 years than after the 10 years.

(3). The service time of the RC subsea tunnel decreases rapidly as the gel/capillary pores radius (r^{peak}) increases on a non-linear curve. Due to the high requirements on the service life of undersea tunnels, the gel/capillary pores radius (r^{peak}) of the concrete is recommended to be controlled below 1×10^{-7} m by adding admixtures. In this case, by appropriately increasing the thickness of the protective layer, the service time of the RC subsea tunnel can be effectively extended.

(4). In the diffusion zone, the concentration of chloride ions significantly increases with increasing gel/capillary pores radius (r^{peak}), and the effects of gel/capillary pores radius (r^{peak}) on the migration process of chloride ions in the concrete lining of subsea tunnels is more sensitive the initial saturation (S_e) than the vG parameters (a and m). Furthermore, with the increase of gel/capillary pores radius (r^{peak}), the sensitivity of the chloride ion migration to gel/capillary pores radius under different chloride binding conditions is increasing.

Declarations

Author Contribution

Yafen Zhang: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing-Original draft, Writing - Review & Editing, Funding acquisition
Ruonan Liu: Validation, Investigation, Resources, Data Curation
Ruicheng Zhang: Validation, Investigation, Data Curation
Yulong Zhu: Conceptualization, Methodology, Writing-Original draft, Writing - Review & Editing, Project administration, Funding acquisition

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Figures

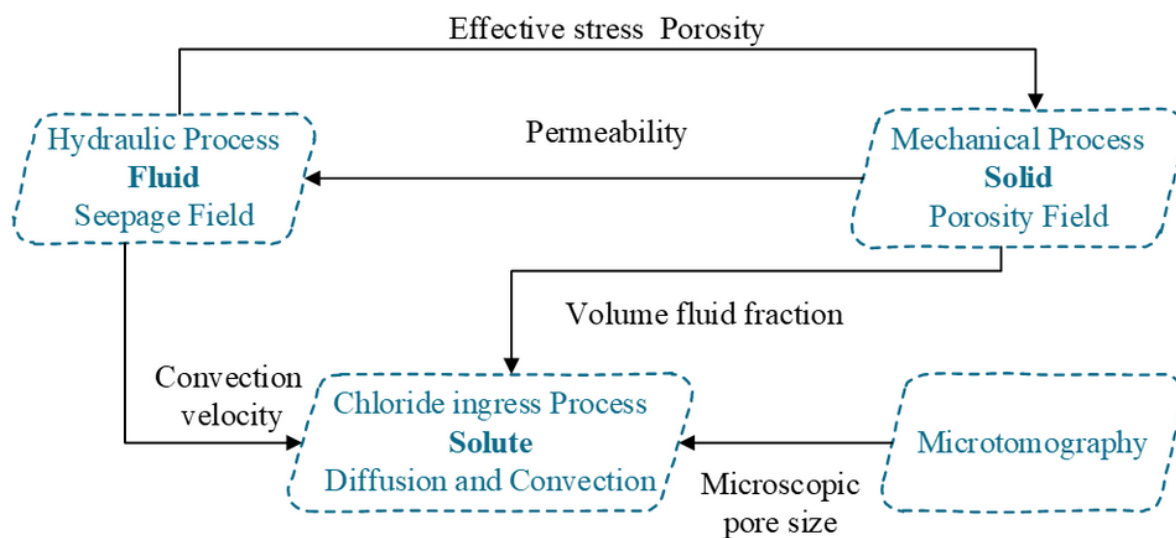


Figure 1

Schematic diagram of H-M-C coupling model.

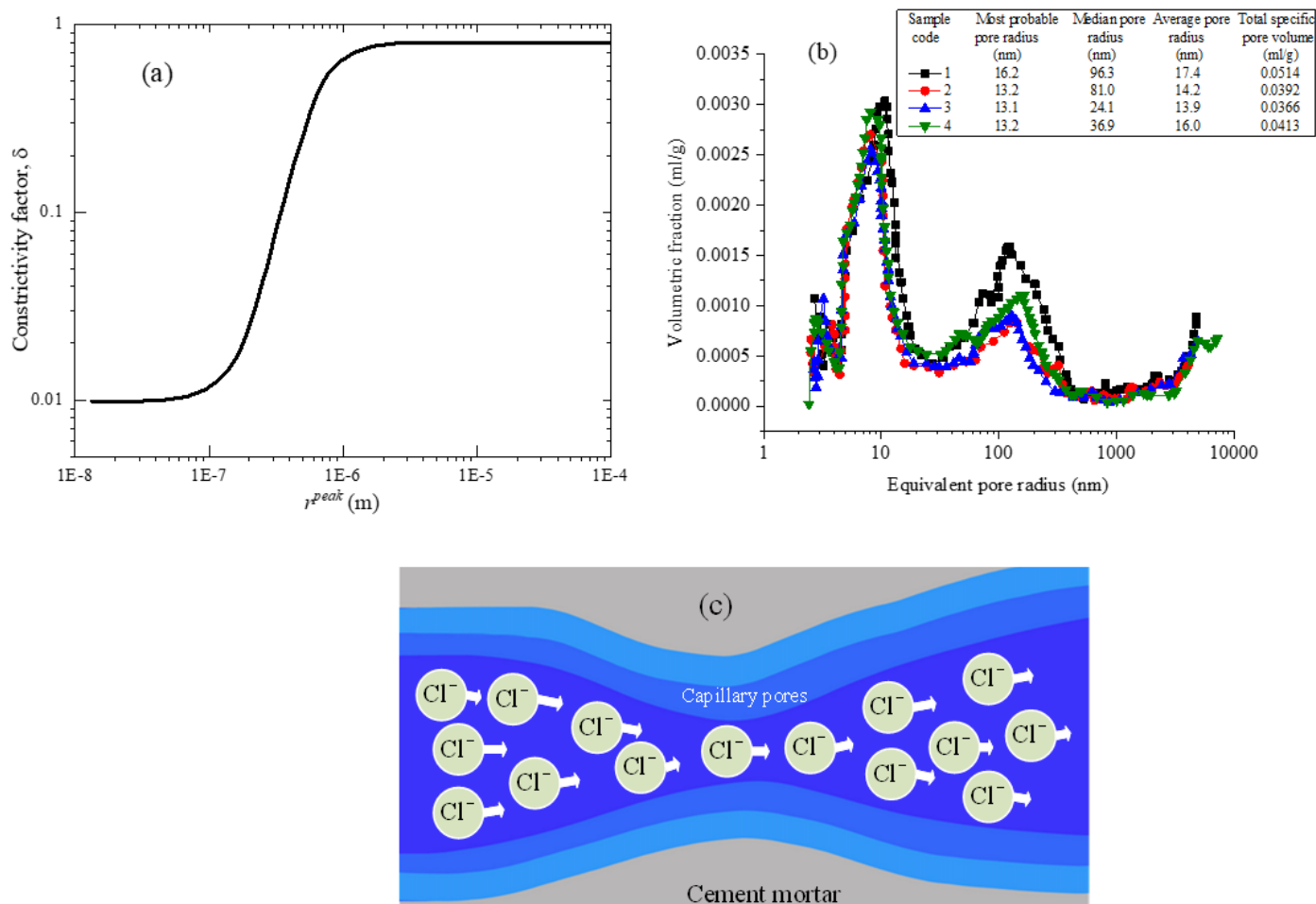


Figure 2

(a). the relationship between r^{peak} and constrictivity; (b). the relationship between pore radius and density of pore volume; (c). the porosity diagram (adopted from Gong et al. (2014), Wang et al. (2022), and Nakarai et al. (2006)).

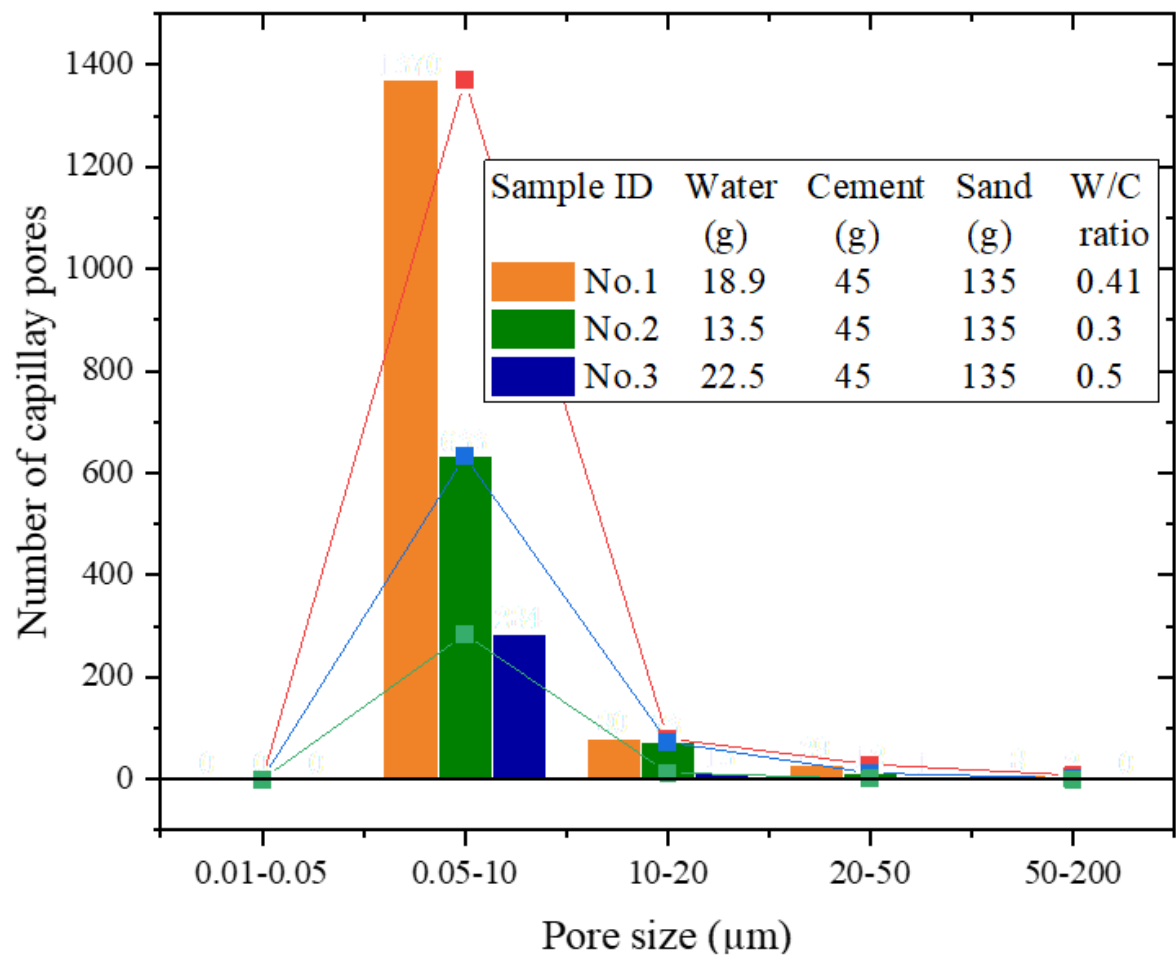


Figure 3

Number of gel/capillary pores with different sizes (Gargepuram et al., 2021).

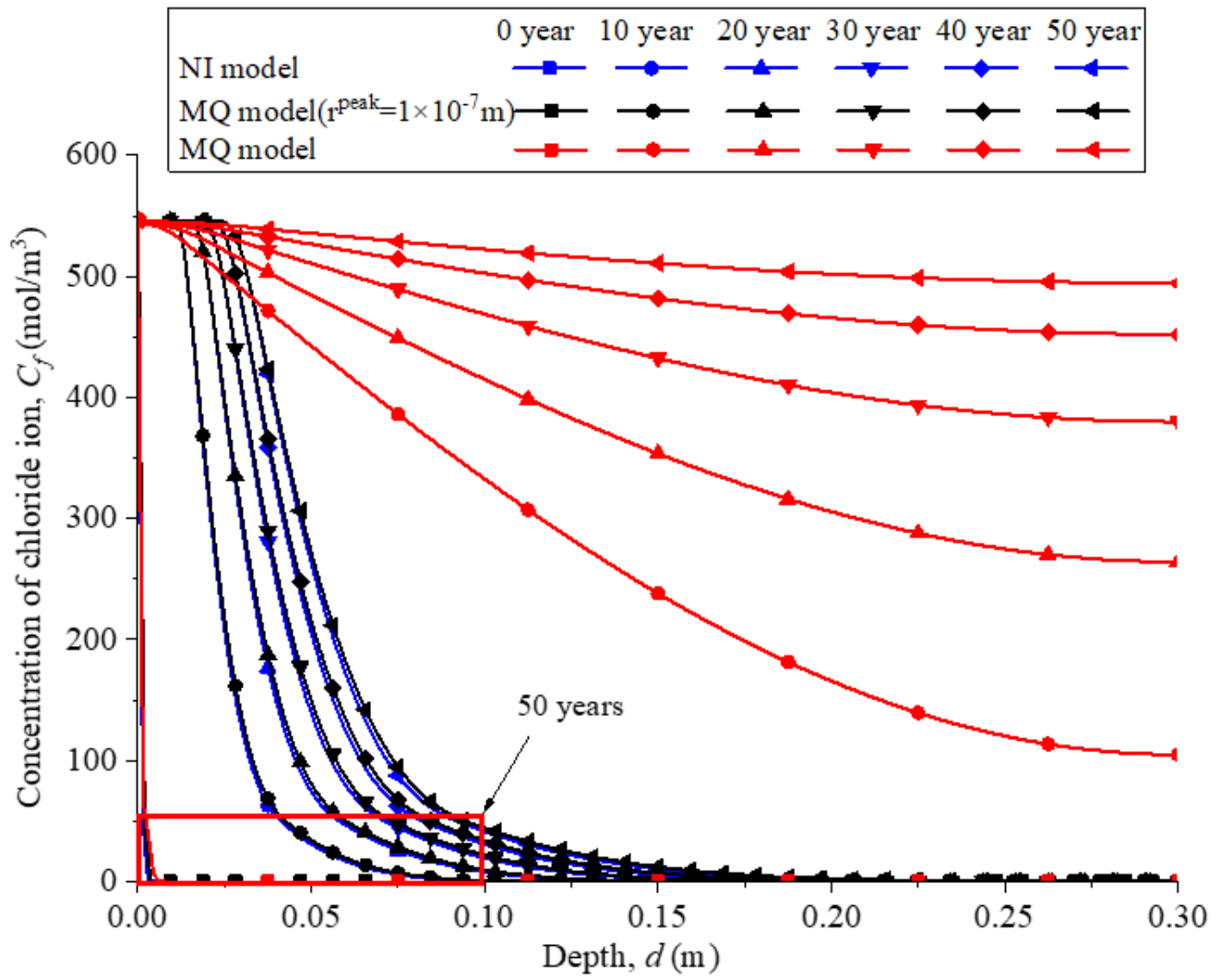


Figure 4

Chloride ions with year for different models with $r^{peak} = 1 \times 10^{-7}$ m.

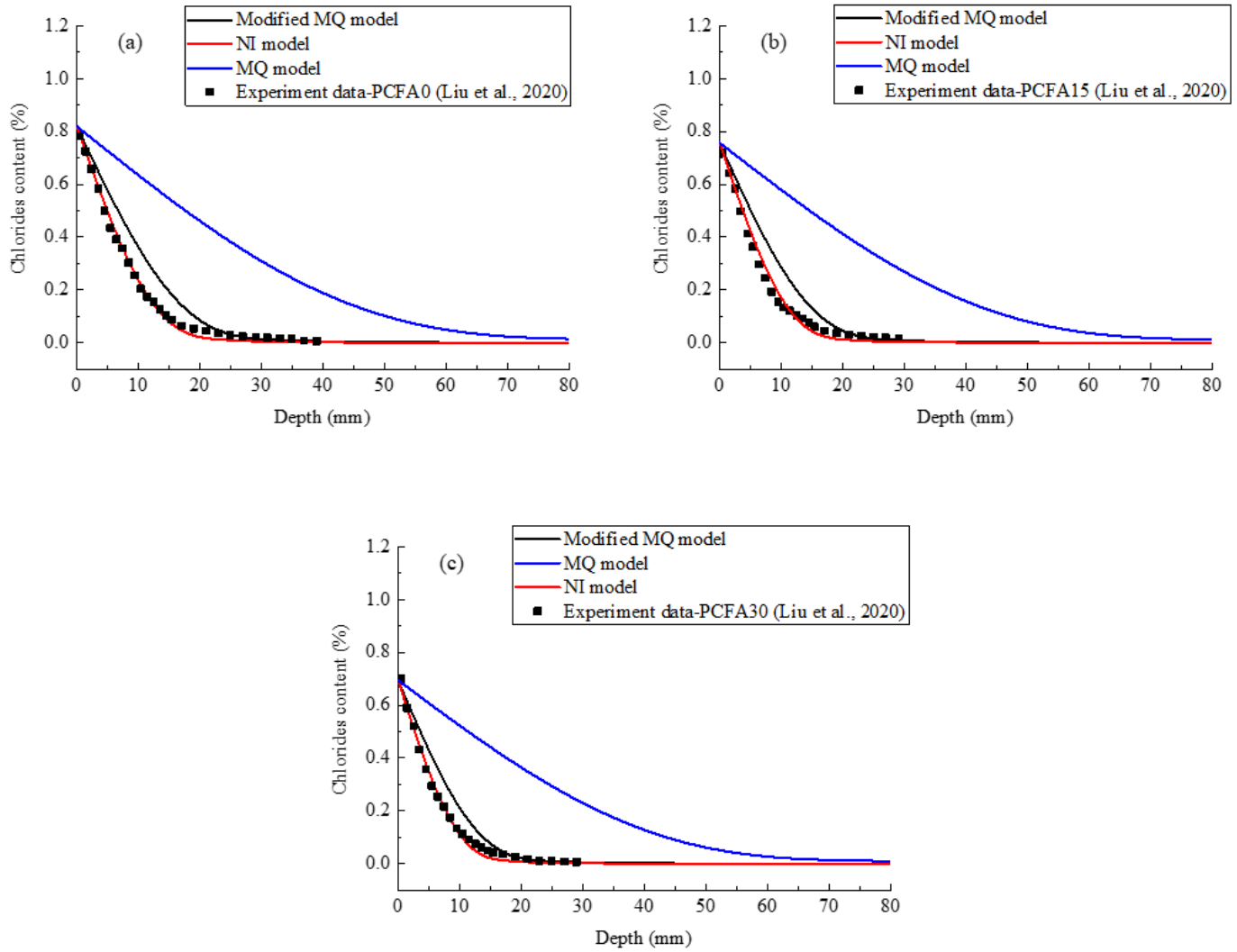


Figure 5

Numerical and experimental results (adopted from Liu et al. (2020)) of distribution of chloride content (by mass %) in the concrete. (a). plain concrete with 0% fly ash by weight (PCFA0); (b). plain concrete with 15% fly ash by weight (PCFA15); (c). plain concrete with 30% fly ash by weight (PCFA30).

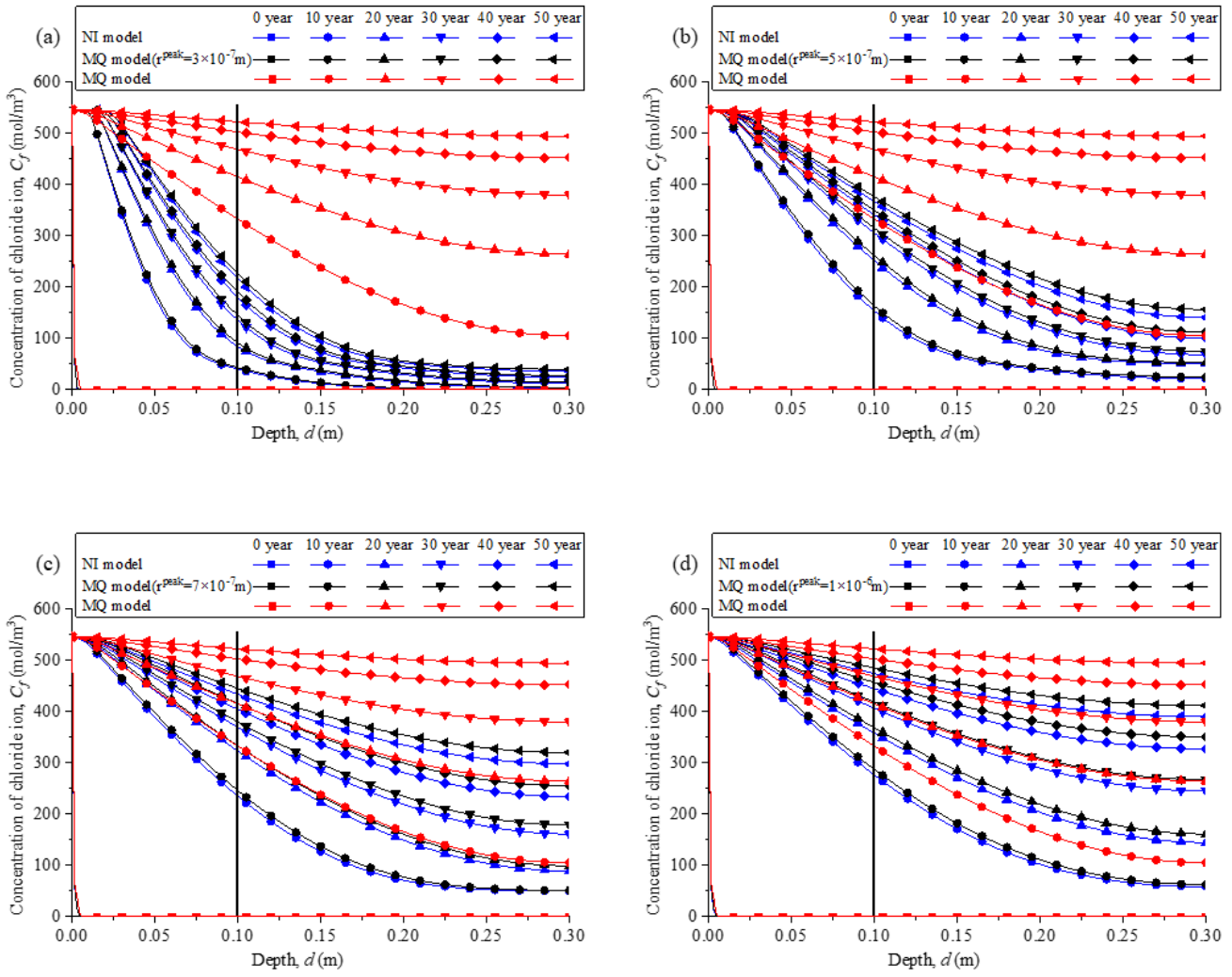


Figure 6

Concentration of chloride ions calculated by the three models with different gel/capillary pores radius (a). $r^{peak}=3\times10^{-7}$ m; (b). $r^{peak}=5\times10^{-7}$ m; (c). $r^{peak}=7\times10^{-7}$ m; (d). $r^{peak}=1\times10^{-6}$ m.

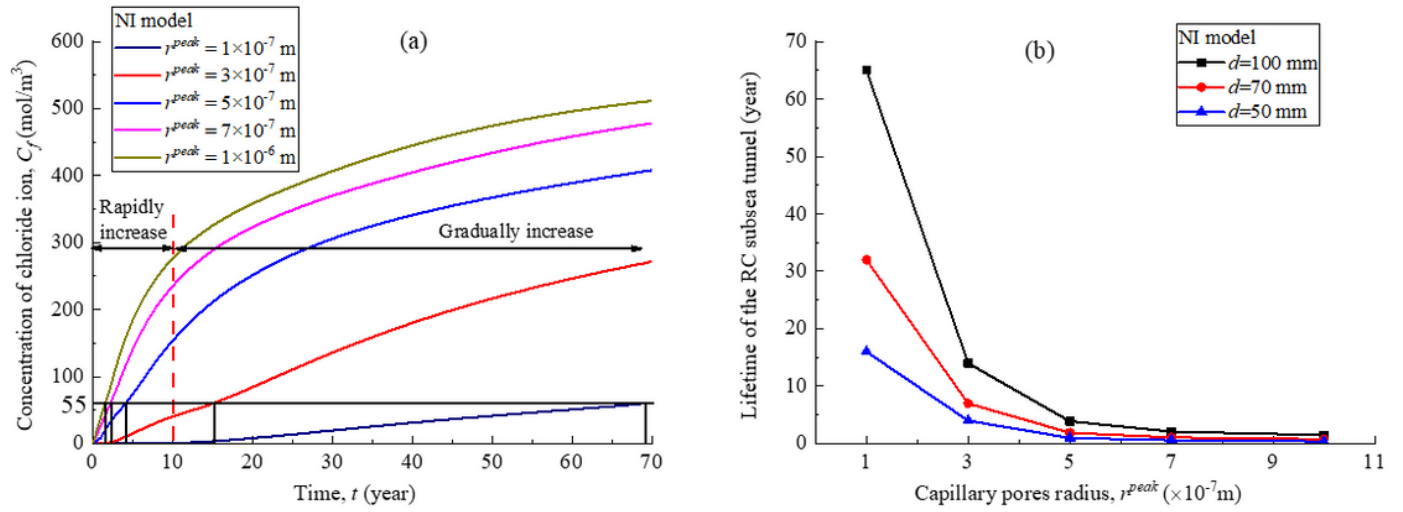


Figure 7

Chloride ions with year for NI model with different gel/capillary pores radius (r^{peak}). (a). time-dependent concentration of chloride ions at the location of the steel bar ($d=100$ mm), and (b). service time of the RC subsea tunnel with different gel/capillary pores radius (r^{peak})

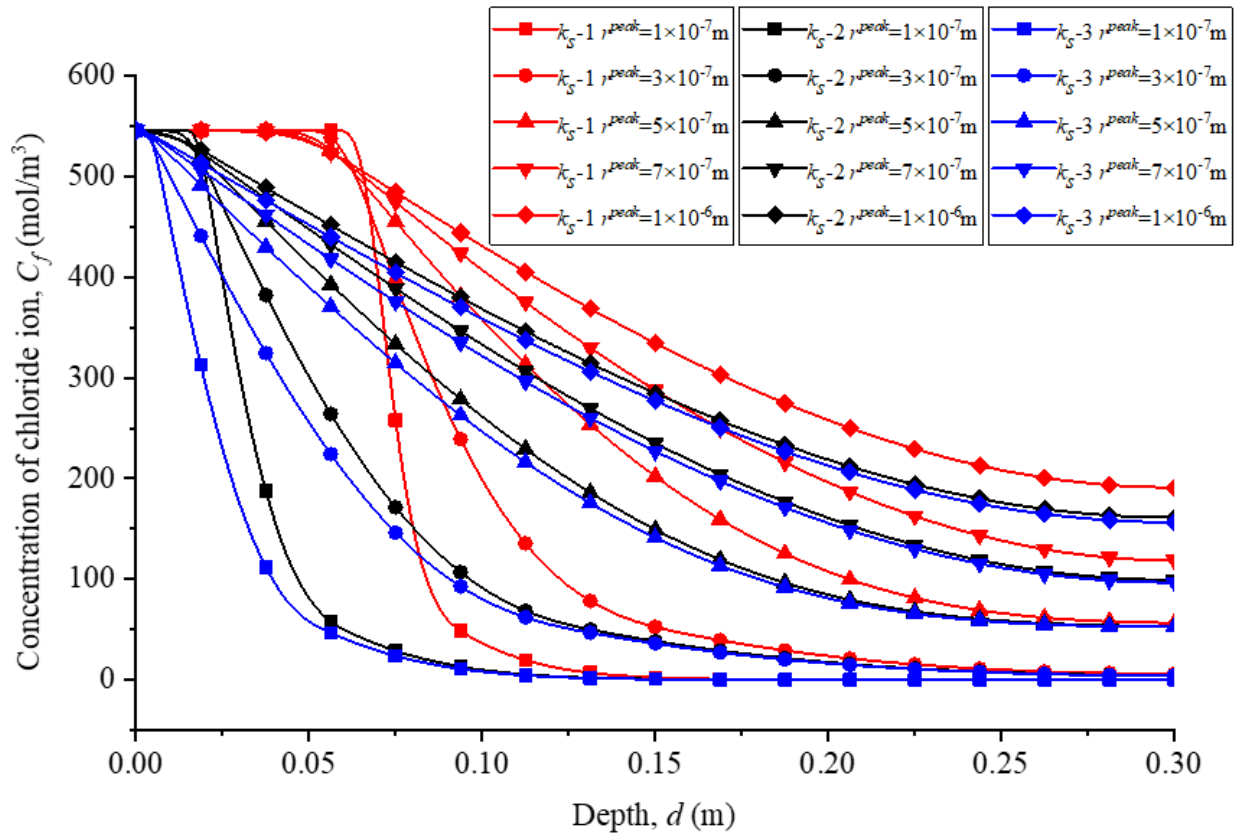


Figure 8

Concentration profiles of chloride ions with different gel/capillary pores radius (r^{peak}) and saturated permeability in 20 years.

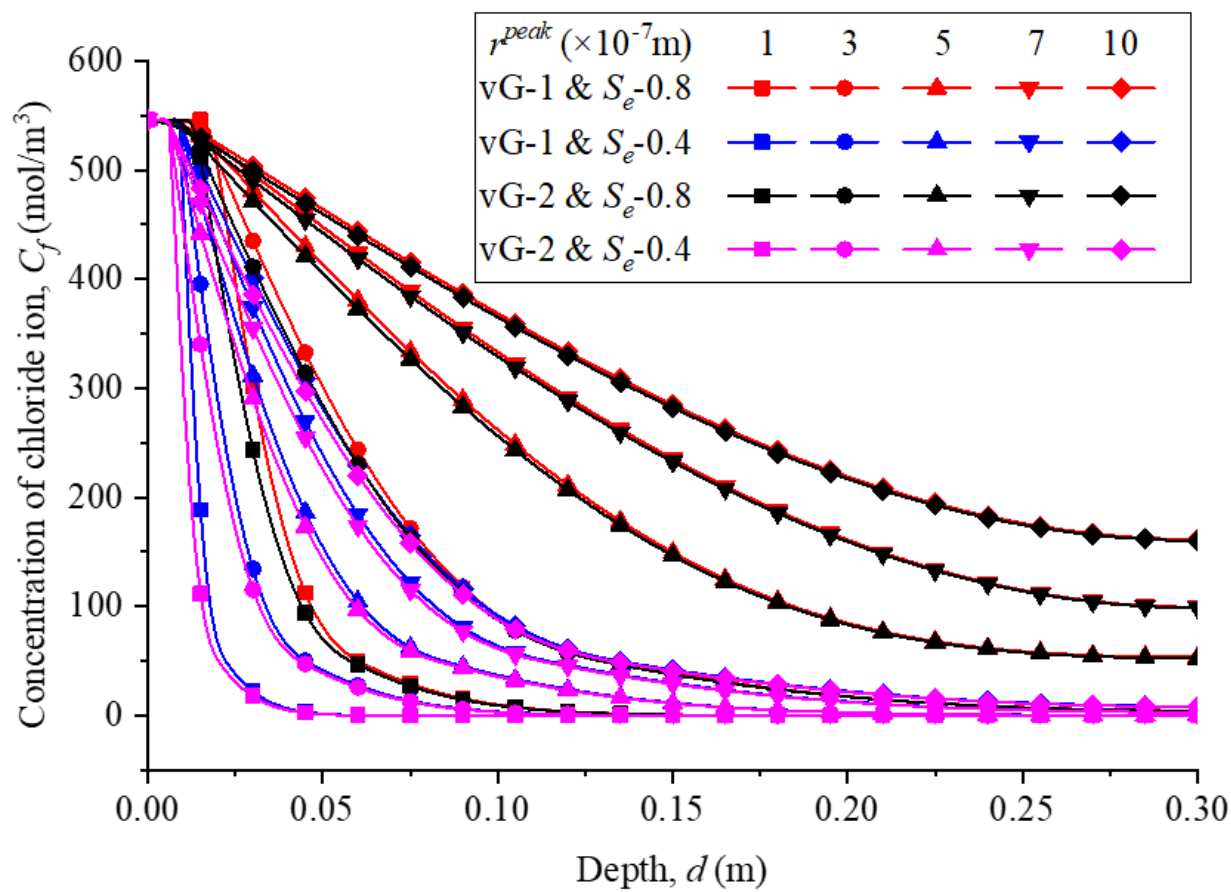


Figure 9

Concentration profiles of chloride ions with different gel/capillary pores radius (r^{peak}), van Genuchten (vG) parameters, and initial saturations in 20 years.

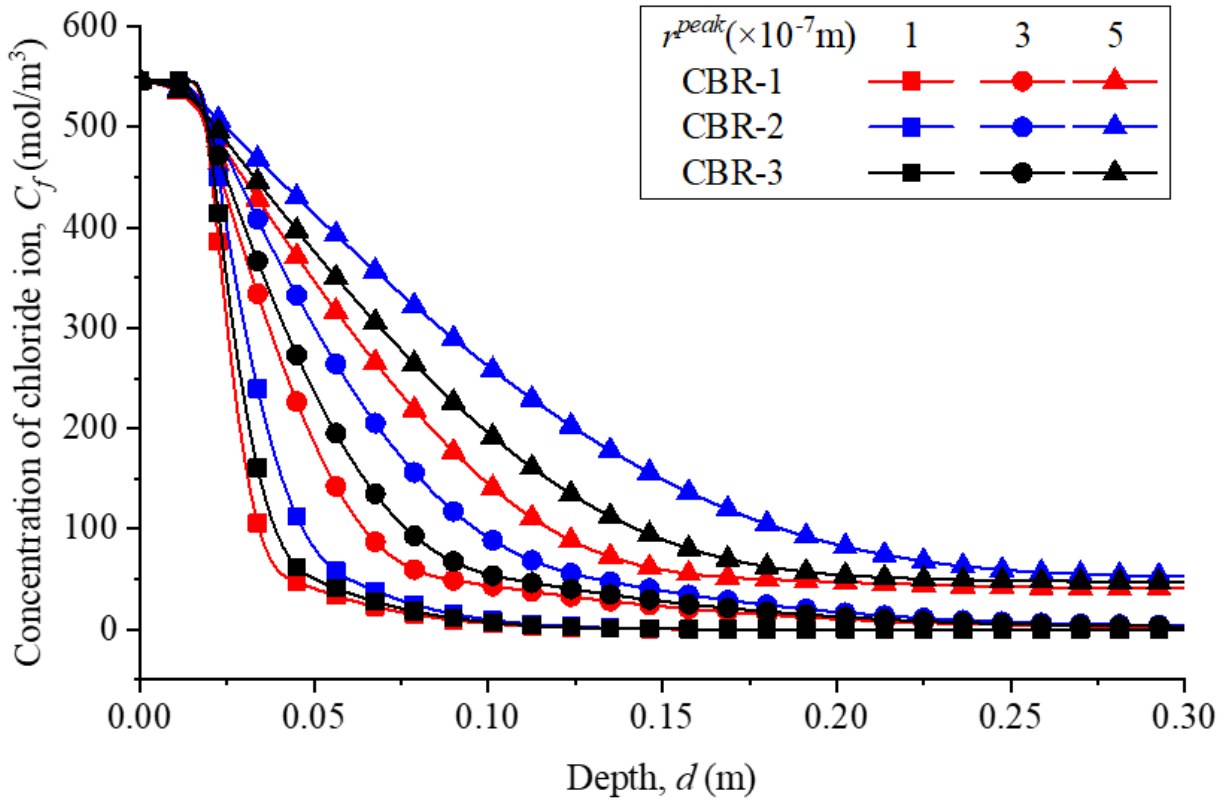


Figure 10

Concentration profiles of chloride ions with different gel/capillary pores radius (r^{peak}) and CBR in 20 years.